noc25_ee134

Mock Exam 2 | Course ID: noc25_ee134

Q1. What does EVSE stand for in the context of electric vehicle charging?

A) Electric Vehicle Safety Equipment
B) Electric Vehicle Supply Equipment
C) Energy Voltage Supply Electronics
D) Electronic Vehicle Service Engine

Source: -GZyYmm0B2M | Difficulty: Easy

Q2. What is the primary purpose of the DC-Link Capacitor in a three-phase AC-DC converter?

A) To increase the switching frequency of the converter.
B) To provide galvanic isolation between the AC and DC sides.
C) To smooth the DC voltage and absorb ripple currents.
D) To control the power factor at the AC input.

Source: -dgKwl8Qoqs | Difficulty: Easy

Q3. In the context of soft switching, what is Zero Voltage Switching (ZVS)?

A) A method where the current through the switch is zero before it turns on.
B) A method where the voltage across the switch is driven to zero before the current begins to rise during turn-on.
C) A technique to eliminate all conduction losses in a switch.
D) A method where the switching frequency is reduced to zero to minimize losses.

Source: -kxjMKhO51M | Difficulty: Easy

Q4. What is the primary motivation for operating DC-DC converters at high switching frequencies?

A) To increase the converter's output voltage.
B) To reduce the size and weight of passive components like transformers and inductors.
C) To decrease conduction losses in the switches.
D) To simplify the control loop design.

Source: -xXPns3RW-g | Difficulty: Easy

Q5. What is the characteristic waveform of the input current drawn from the AC source by a diode bridge rectifier with a capacitive filter?

A) A pure sine wave in phase with the voltage.
B) A square wave with high harmonic content.
C) A triangular waveform synchronized with the voltage peaks.
D) Sharp, narrow pulses, described as "peaky".

Source: 1Ae4n3hWxwY | Difficulty: Easy

Q6. What is a key advantage of a full-bridge DC-DC converter regarding switch voltage stress when compared to a push-pull converter?

A) The switches in a full-bridge converter block half the input voltage (Vin/2).
B) The switches in a full-bridge converter only need to block the input voltage (Vin).
C) The switches in a full-bridge converter block twice the input voltage (2Vin).
D) The switches in a full-bridge converter experience zero voltage stress.

Source: 38D14JN-3wg | Difficulty: Easy

Q7. What is the primary advantage of using the DQ rotating reference frame for controlling three-phase AC-DC converters?

A) It increases the switching frequency of the converter.
B) It transforms AC quantities into DC quantities under steady-state, simplifying controller design.
C) It eliminates the need for an output capacitor.
D) It physically decouples the three phases of the AC source.

Source: 3TF3ZUAokeo | Difficulty: Easy

Q8. What is the primary purpose of the Open Charge Point Protocol (OCPP)?

A) To standardize the physical shape of charging connectors.
B) To define the voltage and power levels for DC fast charging.
C) To enable interoperability and standardized communication between EVSEs and Central Management Systems (CMS).
D) To regulate the price of electricity at public charging stations.

Source: UAKL2GPPj0U | Difficulty: Easy

Q9. In Sinusoidal Pulse Width Modulation (SPWM), what parameter directly controls the amplitude of the fundamental component of the output voltage?

A) The carrier frequency (fc).
B) The DC link voltage (Vdc).
C) The modulation index (m).
D) The dead time.

Source: VP2T2QgDMws | Difficulty: Easy

Q10. In a DC charging scenario, where does the AC-to-DC power conversion take place?

A) Inside the vehicle's onboard charger.
B) Within the vehicle's Battery Management System (BMS).
C) Outside the vehicle, within the DC EVSE (off-board charger).
D) At the grid substation before reaching the charging station.

Source: a-fvKWBRT_g | Difficulty: Easy

Q11. What is the primary difference between AC and DC charging in terms of where the power conversion from AC to DC occurs?

A) In AC charging it happens in the EVSE, while in DC charging it happens inside the vehicle.
B) Power conversion happens inside the vehicle for both AC and DC charging.
C) In AC charging it happens inside the vehicle's onboard charger, while in DC charging it happens within the EVSE.
D) Power conversion is not needed for DC charging.

Source: -GZyYmm0B2M | Difficulty: Medium

Q12. According to the energy equivalence principle, how is the minimum required DC-link capacitance (C_min) determined?

A) It is directly proportional to the nominal DC voltage (V_0) and voltage ripple (ΔV_0).
B) It is inversely proportional to the maximum power variation (ΔP_{max}) and response time (T_d).
C) It is calculated as ($\Delta P_{max} * T_d$) / (2 * V_0 * ΔV_0).
D) It is based solely on the RMS current flowing through the capacitor.

Source: -dgKwI8Qoqs | Difficulty: Medium

Q13. In a conventional half-bridge converter with a constant inductive load, what factor determines which of the two switches achieves ZVS turn-on?

A) The value of the DC input voltage (V_in).
B) The switching frequency (F_sw).
C) The direction of the load current (I0).

D) The length of the dead time.
Source: -kxjMKhO51M | Difficulty: Medium

Q14. In transformer design, what is the relationship between the Peak AC Flux Density (deltaB) and the two primary loss components?

A) Both core loss and copper loss increase as deltaB increases.
B) Core loss increases with deltaB, while copper loss decreases with deltaB.
C) Both core loss and copper loss decrease as deltaB increases.
D) Core loss decreases with deltaB, while copper loss increases with deltaB.
Source: 24SWNw-ePds | Difficulty: Medium

Q15. What is the primary reason for introducing an air gap into the core of a filter inductor?

A) To increase the inductance value for the same number of turns.
B) To reduce the physical size and weight of the core.
C) To increase the inductor's saturation current capability.
D) To eliminate all copper losses in the winding.
Source: 6X9uRXj0MZQ | Difficulty: Medium

Q16. According to the Empirical Steinmetz Formula, how does eddy current loss generally relate to operating frequency (f) and peak AC flux density (ΔB)?

A) It is proportional to f and ΔB.
B) It is proportional to f^2 and ΔB^2.
C) It is inversely proportional to f and ΔB.
D) It is proportional to f and ΔB^2.
Source: CjNOVQf9HiA | Difficulty: Medium

Q17. In the CCS2 protocol, what is the role of the `EVSE processing` field within response messages like `ChargeParameterDiscoveryRes`?

A) It indicates the total processing time taken by the EVSE.
B) It signals whether the EVSE is still determining parameters ('ongoing') or has completed the task ('finished').
C) It specifies the CPU load of the Supply Equipment Communication Controller (SECC).
D) It provides an error code if the processing fails.
Source: EGTWtv4aqGg | Difficulty: Medium

Q18. What is the primary cause of the "double line frequency ripple" on the output capacitor of a single-phase boost PFC converter operating at unity power factor?

A) The high-frequency switching of the boost converter's main switch.
B) The interaction between the input power, which has a DC and a $2\omega t$ component, and the constant power drawn by the load.
C) The reverse recovery characteristics of the rectifier diodes.
D) The parasitic inductance of the circuit wiring.
Source: LGd3GukayGs | Difficulty: Medium

Q19. Why is a high-frequency transformer preferred over a low-frequency (50/60 Hz) transformer for providing galvanic isolation in an EVSE?

A) Low-frequency transformers cannot provide sufficient galvanic isolation.
B) High-frequency operation allows for a significantly smaller and lighter transformer core for the same power level.
C) High-frequency transformers have lower core losses at the same flux density.
D) Low-frequency transformers are incompatible with DC-DC converter topologies.
Source: TlKIeTe3Fy0 | Difficulty: Medium

Q20. What is the key advantage of using a JSON/WebSocket-based OCPP implementation over a SOAP/XML-based one regarding scalability?

A) JSON messages are encrypted by default, while SOAP messages are not.
B) SOAP requires a dedicated physical wire for each EVSE.
C) WebSockets allow multiple EVSEs to communicate over a single persistent connection without needing unique public IP addresses for each.
D) JSON is only compatible with high-power DC chargers, which are more scalable.
Source: WUYG4Eko3rU | Difficulty: Medium

Q21. What is the most significant advantage of the Totem Pole PFC converter over the standard bridgeless PFC converter?

A) It uses fewer active switches.
B) It completely eliminates the need for an inductor.
C) Its diodes (D1, D2) switch at the low line frequency, drastically reducing reverse recovery losses.
D) It can operate without a closed-loop control system.
Source: X4acp5BiIFs | Difficulty: Medium

Q22. What is the critical condition that must be met to achieve Zero Voltage Switching (ZVS) turn-on for the leading leg switches in a PSFB converter?

A) The dead time must be exactly zero.
B) The leakage inductance must be minimized to zero.
C) The dead time must be greater than the time required for the primary current to discharge the switch's output capacitance (e.g., > 2C*Vin / (N*I0)).
D) The switching frequency must be lower than the line frequency.
Source: YINqDUxMemQ | Difficulty: Medium

Q23. Why is the simplified power balance approach, which works for single-phase converters, inadequate for accurately sizing the DC-link capacitor in a three-phase AC-DC converter?

A) Because three-phase converters do not require a DC-link capacitor.
B) Because the power in a three-phase system is always zero.
C) Because it neglects the high-frequency switching harmonics which are the primary cause of capacitor ripple current in a balanced three-phase system.
D) Because three-phase converters operate at a much lower voltage.
Source: -dgKwI8Qoqs | Difficulty: Hard

Q24. According to the lecture, what is the correct structure of a JSON-based OCPP 1.6J "Boot Notification Request" message?

A) {"Action": "BootNotification", "Payload": {"vendor": "X", "model": "Y"}}
B) [2, "unique_id", "BootNotification", {"chargePointVendor": "X", "chargePointModel": "Y"}]
C) <BootNotification><vendor>X</vendor><model>Y</model></BootNotification>
D) ["BootNotification", "unique_id", 2, {"chargePointVendor": "X", "chargePointModel": "Y"}]
Source: 4TnD50QrkVg | Difficulty: Hard

Q25. During an AC Type-2 charging sequence, how does the EVSE detect that the EV is ready for charging, prompting the transition to State C2?

A) The EV sends a high-level communication message over the CAN bus.
B) The EVSE detects a change in the Proximity Pilot (PP) pin resistance.
C) The EV closes its internal S2 switch, which changes the voltage divider on the Control Pilot (CP) line, causing the positive peak voltage to drop from +9V to +6V.
D) The EVSE measures an increase in the PWM signal's duty ratio to above 85%.
Source: 6-xQQwjzPIc | Difficulty: Hard

Q26. During the CCS2 charging sequence, what specific mechanism prevents a large inrush current when the EV closes its contactor switch?

A) The EVSE limits the current to 2A and matches its output voltage to the battery voltage within a 20V tolerance during the pre-charge phase.
B) The EV uses a large internal resistor to absorb the initial current surge.
C) The charger plug is not fully locked until the voltages are equalized.
D) The high-level communication protocol negotiates a zero-current start.

Source: EGTWtv4aqGg | Difficulty: Hard

Q27. In the design of the inner current loop for a three-phase AC-DC converter, what is the primary purpose of using pole-zero cancellation in the PI controller design?

A) To increase the overall gain of the system to infinity.
B) To cancel the pole of the plant (L-Rs impedance), simplifying the system dynamics and making controller gain calculation more straightforward.
C) To eliminate the need for a current sensor.
D) To force the converter to operate in discontinuous conduction mode.

Source: KmgcXbLuwno | Difficulty: Hard

Q28. As described in the lecture, what is the specific mechanism that causes a negative current spike during the zero-crossing transition in a Totem Pole PFC converter?

A) The output capacitor discharges back into the grid.
B) The body diodes of the low-frequency switches fail to conduct.
C) When S4 is on and vs is near zero, the brief turn-on of S1 applies a large negative voltage (vs - Vo ≈ -Vo) across the inductor, causing a steep negative current slope.
D) The control loop overcompensates by demanding a large negative current to maintain power factor.

Source: RxJCQteGJQA | Difficulty: Hard

Q29. Explain the two-level optimization strategy for minimizing total losses in a high-frequency transformer design.

A) First, optimize the core material for low loss, then optimize the wire material.
B) First, minimize core loss by choosing a low ΔB, then minimize copper loss by choosing a large wire.
C) First, optimize the window area allocation (γ_j) to minimize total copper loss, then optimize the peak flux density (ΔB) to minimize the sum of core and copper losses.
D) First, optimize the switching frequency for minimum loss, then optimize the turns ratio.

Source: WwzCJVUrYNY | Difficulty: Hard

Q30. According to the lecture, what are the three main design criteria for the inner current loop controller (Cc(s)) in a single-phase boost PFC converter?

A) Low DC gain, bandwidth equal to switching frequency, and zero phase margin.
B) High gain at twice the line frequency (2fs), bandwidth significantly less than switching frequency (fsw), and sufficient phase/gain margins.
C) High gain at the switching frequency, bandwidth equal to the line frequency, and a gain margin of zero.
D) Low gain at all frequencies, bandwidth greater than switching frequency, and a phase margin of 90 degrees.

Source: kGh_kDgqomA | Difficulty: Hard

Answer Key

1: B	2: C	3: B	4: B	5: D
6: B	7: B	8: C	9: C	10: C
11: C	12: C	13: C	14: B	15: C
16: B	17: B	18: B	19: B	20: C
21: C	22: C	23: C	24: B	25: C
26: A	27: B	28: C	29: C	30: B